Course: CS 108 Statistics

Team Number: Team 5

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Dataset: Bike Sharing Dataset

1. Dataset Description

1. The dataset comes from the UCI Machine Learning Repository and is called the Bike Sharing Dataset. It contains bike rental data from the Capital Bikeshare system for 2011 and 2012, together with weather and seasonal information.

2. One observation in day.csv represents one day of bike-sharing activity.

3. The dataset describes the daily number of bike rentals in the Capital Bikeshare system during 2011–2012, together with calendar, weather, and seasonal information.

4-5. The day.csv dataset contains 731 observations and 16 variables. The UCI description lists 17 variables because it describes both day.csv and hour.csv. The variable hr is available only in hour.csv, so it is not included in our dataset.

The variables are:

Variable	**Type**	**Description**	**Units**	**Missing Values**
instant	Integer	Record index	No unit	no
dteday	Date	Date	Date (YYYY-MM-DD)	no
season	Categorical	Season (1: spring, 2: summer, 3: fall, 4: winter)	No unit	no
yr	Categorical	Year (0: 2011, 1: 2012)	No unit	no
mnth	Categorical	Month (1 to 12)	No unit	no
holiday	Binary	Whether the day is a holiday	No unit	no
weekday	Categorical	day of the week	No unit	no

workingday	Binary	if day is neither weekend nor holiday is 1, otherwise is 0	No unit	no
weathersit	Categorical	- 1: Clear, Few clouds, Partly cloudy, Partly cloudy	No unit	no
temp	Continuous	Normalized temperature in Celsius. The values are derived via (t-t_min)/(t_max-t_min), t_min=-8, t_max=+39 (only in hourly scale)	C	no
atemp	Continuous	Normalized feeling temperature in Celsius. The values are derived via (t-t_min)/(t_max-t_min), t_min=-16, t_max=+50 (only in hourly scale)	C	no
hum	Continuous	Normalized humidity. The values are divided to 100 (max)	Normalized value; humidity divided by 100	no
windspeed	Continuous	Normalized wind speed. The values are divided to 67 (max)	Normalized value; wind speed divided by 67	no
casual	Integer	count of casual users	Users	no
registered	Integer	count of registered user	Users	no
cnt	Integer	count of total rental bikes including both casual and registered	Rentals (casual + registered)	no

Source: UCI Machine Learning Repository, *Bike Sharing Dataset*

https://archive.ics.uci.edu/dataset/275/bike+sharing+dataset

2. Data Preparation

The dataset was imported into R and checked for missing values, errors, inconsistencies, and unusual observations. Categorical variables such as season, weather situation, weekday, and

working-day status were converted to factors with meaningful labels. Potential outliers were not automatically removed and were investigated separately.

3. Categorical Variables

Season and weekday are approximately balanced across the dataset. Clear/Few Clouds is the most common weather condition, occurring on about 63% of days, while Light Snow/Rain is rare at about 3%. Approximately 68% of observations are working days.

The categorical comparisons also showed that weather conditions vary somewhat across seasons. Working-day status is strongly related to weekday: weekends are non-working days, while most weekdays are working days.

4. Numerical Variables

Temperature and total bike rentals have relatively balanced distributions. Mean daily rentals are approximately 4504, with a median of 4548, showing substantial day-to-day variability.

Humidity contains two potential low outliers. Wind speed is right-skewed and contains several potential high outliers. These observations were retained for further investigation rather than automatically removed.

5. Numerical Variables by Categories

Daily bike rentals differ across seasons and weather conditions. Rental activity is generally higher in Summer and lower in Winter. Less favorable weather, particularly Light Snow/Rain, is generally associated with lower rental activity.

The distributions also differ in spread and shape across groups, indicating that changes are not limited only to their averages.

6. Relationships Between Numerical Variables

Temperature and total bike rentals have a positive correlation of approximately 0.6275, while humidity has only a weak negative correlation with rentals of approximately −0.1007.

The temperature-rental relationship also differs by season. It is strongest in Winter (0.669), positive in Spring (0.477) and Fall (0.398), and almost absent in Summer (−0.033). Therefore, the overall relationship is not equally strong within every season.

7. Distributional Analysis

Normalized temperature was investigated as a possible Normal distribution. Although its mean and median are very close, the overall QQ plot shows systematic departures from the Normal reference line, especially in the tails. Therefore, the overall temperature distribution is not well described by a single Normal distribution.

Temperature distributions also differ considerably across seasons in their centers, variability, and QQ-plot patterns. These seasonal differences help explain the shape of the combined temperature distribution.

8. Integrated Exploratory Analysis

Three integrated questions were investigated. First, both casual and registered rentals vary by season, but seasonal differences are especially noticeable for casual users.

Second, the positive relationship between temperature and total rentals remains similar across weather conditions, with correlations of 0.622 for Clear/Few Clouds, 0.645 for Mist/Cloudy, and 0.607 for Light Snow/Rain.

Third, working-day status is associated differently with casual and registered rentals. Casual rentals are substantially higher on non-working days, while registered rentals are higher on working days. This suggests different usage patterns for the two types of users.

9. Outliers and Unusual Observations

The boxplot method identified 2 potential outliers for humidity, 13 for wind speed, and 44 high outliers for casual rentals. Removing the humidity and wind-speed outliers produced only small changes in their means and medians, so their influence was limited. The casual-rental outliers had a larger effect: the mean decreased from 848.2 to 732.0 when they were excluded, while the median decreased from 713 to 674.

There was no clear evidence that these observations were data-entry errors. They may represent legitimate unusual weather conditions or days with particularly high casual bike usage. Therefore, the potential outliers were retained in the analysis.

10. Conclusions

The analysis revealed clear patterns in bike-sharing activity. Rental activity is generally higher in Summer and lower in Winter, and less favorable weather conditions are associated with lower usage. Temperature has a positive relationship with total rentals, although the strength of this relationship differs across seasons.

An important finding was the difference between casual and registered users. Casual rentals are substantially higher on non-working days, while registered rentals are higher on working days. Seasonal differences are also particularly noticeable for casual users.

The overall normalized temperature distribution was not well described by a single Normal distribution, and its distribution differed across seasons. Potential outliers were also identified, but there was no evidence that they were errors, so they were retained.

Some limitations should be considered. Light Snow/Rain contains relatively few observations, and the dataset does not record the purpose of individual trips. Other variables may also contribute to the observed patterns. Therefore, the results describe statistical patterns and associations and should not be interpreted as evidence of causation.

References

Dataset: UCI Machine Learning Repository. (n.d.). *Bike sharing dataset*.

https://archive.ics.uci.edu/dataset/275/bike+sharing+dataset

AI tools were used to assist with writing, organization, and clarification of technical tasks during the analysis.